

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)		a = acceleration ω = angular velocity or angular frequency or pulsance x = displacement All 3 correct (1)	1			1		
	(b)	(i)	$\omega = \frac{2\pi}{0.4} (1) [= 15.7 \text{ s}^{-1}]$ $a_{\text{max}} = \omega^2 A = (15.7)^2 \times 0.012 (1)$ $= 2.96 [\text{m s}^{-2}] (1)$	1	1 1		3	2	
		(ii)	One value of a on graph (1) ecf from (b)(i) Any straight line (1) Correct position of line (1)		3		3	2	
	(c)		$x = \boxed{0.012} \cos\left(\boxed{15.7} t + \boxed{\frac{3\pi}{2}}\right) \quad (1 \times 3 - \text{one mark for each box})$ (alternative for angle: $-\frac{\pi}{2}$) Accept 5π for 15.7 . ecf on ω		3		3	1	
	(d)	(i)	Example e.g. microwave ovens or swing (1) Oscillator and driving force named e.g. water molecules and microwaves or swing and person pushing (1)	2			2		
		(ii)	Example and consequence e.g. bridge and falling or something in the dashboard and buzzing (1) Driving force e.g. wind / soldiers marching or engine (1) Resonance explained i.e. both frequencies are the same (1)	3			3		
			Question 3 total	7	8	0	15	5	0